

Q1). “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?”

Bayesian Network – Machine Failure

A **Bayesian Network** is a graphical model used to represent relationships between different variables using probability.

For machine failure, we can take sensor values such as **temperature, vibration, pressure and motor current**. These values are connected to the machine failure node.

When the machine is running, the system continuously receives sensor data. The Bayesian Network uses this data to calculate the **probability of machine failure**.

For example, if the temperature and vibration are very high, the probability of failure may increase. If the probability is high, the system can give an alert and maintenance can be performed before the machine breaks down.

So, Bayesian Networks are useful for **predictive maintenance and early failure detection**.

Main point:

Sensor data → Bayesian Network → Failure probability → Alert

Q2. “Imagine you are asked in an interview to design an image denoising solution. How would

Markov Random Fields help model neighborhood pixel dependencies?”

A **Markov Random Field**, or MRF, is an undirected graphical model used to represent relationships between neighboring variables.

In an image, each pixel is connected to its nearby pixels. Normally, neighboring pixels have similar values.

If one pixel has a very different value because of noise, the MRF compares it with its neighboring pixels and helps identify it as noise.

For example, if surrounding pixels have values around 10 or 11, but one pixel has value 99, that pixel may be noise.

Therefore, MRFs can use **neighboring pixel relationships to remove noise and produce a cleaner image.**

Main point:

Pixel → Neighboring pixels → Detect noise → Clean image

Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden

Markov Model to process sequential audio signals?”

A **Hidden Markov Model**, or HMM, is a probabilistic model mainly used for **sequential data**.

Speech is sequential because sounds occur one after another.

In speech recognition, the actual speech sounds are considered hidden states, while the audio signals are observations.

The system first takes the audio and extracts features such as **MFCC**. The HMM then finds the most likely sequence of speech sounds.

For example, when we speak a word, the sounds occur in a particular order. The HMM analyzes this sequence and identifies the most likely word.

The **Viterbi algorithm** can be used to find the most probable sequence of hidden states.

So, HMM is useful for converting **speech signals into words**.

Main point:

Audio → Features → HMM → Recognized word